

Chapter 10: Convex Functions: Variants

Uniform, Strong, and Quasi-convexity in Convex Analysis

Based on Bauschke & Combettes, 2e (2017)

Chapter Outline

- 1 Introduction: Sublinear and Perspective Functions
- 2 Uniform and Strong Convexity
- 3 Quasiconvexity
- 4 Summary and Applications

Perspective and Recession Functions (Examples 10.5-10.6)

Definition

Let $\varphi : \mathcal{H} \rightarrow]-\infty, +\infty]$ be convex and define the **perspective function**:

$$f : \mathbb{R} \times \mathcal{H} \rightarrow]-\infty, +\infty] : (\xi, x) \mapsto \begin{cases} \xi\varphi(x/\xi), & \text{if } \xi > 0; \\ +\infty, & \text{otherwise} \end{cases}$$

Then f is **sublinear**.

Key insight: The perspective function preserves and extends convexity by adding the scale parameter ξ .

Corollary

For a convex function $f \in \Gamma_0(\mathcal{H})$, its **recession function**

$$(\text{rec } f)(y) = \liminf_{x \rightarrow +\infty} \frac{f(x + \alpha y) - f(x)}{\alpha}$$

is **sublinear**.

Example: Perspective of Log Function

- Original function: $\varphi(x) = -\log(x)$ (convex for $x > 0$)
- Perspective: $f(\xi, x) = \xi\varphi(x/\xi) = -\xi \log(x/\xi) = -\xi[\log x - \log \xi] = \xi \log \xi - \xi \log x$
- Properties:
 - Positively homogeneous: $f(t\xi, tx) = tf(\xi, x)$ for $t > 0$
 - Convex in both arguments
 - Used in information geometry and convex optimization

Definition 10.7: Uniform and Strong Convexity

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and $\phi : \mathbb{R}_+ \rightarrow [0, +\infty]$ increasing, vanishing only at 0.

Definition

f is **uniformly convex** with modulus ϕ if:

$$(\forall x \in \text{dom } f)(\forall y \in \text{dom } f)(\forall \alpha \in [0, 1])$$

$$f(\alpha x + (1 - \alpha)y) + \alpha(1 - \alpha)\phi(\|x - y\|) \leq \alpha f(x) + (1 - \alpha)f(y)$$

Definition

If $\phi(t) = (\beta/2) \cdot t^2$ for some $\beta \in \mathbb{R}_{++}$, then f is **strongly convex** with constant β (or modulus of strong convexity).

Hierarchy: Strong convexity $\Rightarrow$ Uniform convexity $\Rightarrow$ Strict convexity $\Rightarrow$ Convexity

Proposition 10.8: Characterizing Strong Convexity

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and $\beta \in \mathbb{R}_{++}$. Then f is strongly convex with constant β if and only if $f - (\beta/2)\|\cdot\|^2$ is convex.

Proof sketch:

- Define $g(x) = f(x) - (\beta/2)\|x\|^2$
- Show that strong convexity of f is equivalent to convexity of g
- Direct consequence of Corollary 2.15

Example 10.9:

- $\|\cdot\|$ is sublinear but not strictly convex (hence not uniformly convex)
- $\|\cdot\|^2$ is strongly convex with constant $\beta = 2$ but NOT positively homogeneous

Computing Exact Moduli: Example 10.10

Example

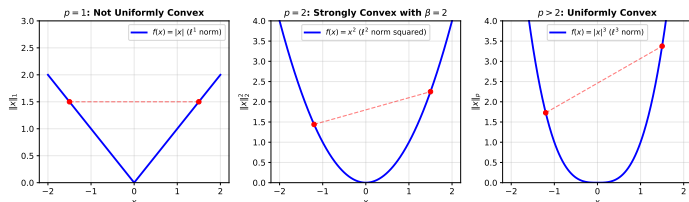
Let $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be increasing and $C \subseteq \mathcal{H}$ bounded and convex. Set

$$f : \mathcal{H} \rightarrow \mathbb{R} : x \mapsto \int_0^{\|x\|} \phi(t) dt$$

Then f is uniformly convex on C with modulus determined by ϕ .

Example applications:

- $p \in]1, +\infty[$ and $\phi(t) = t^{p-1}$: then $\|\cdot\|_p^p$ is uniformly convex
- Particular case $p = 2$: $\phi(t) = t$ gives $\|\cdot\|_2^2$ strongly convex with $\beta = 2$
- Higher p : stronger uniform convexity



Definition 10.11: Exact Modulus of Convexity

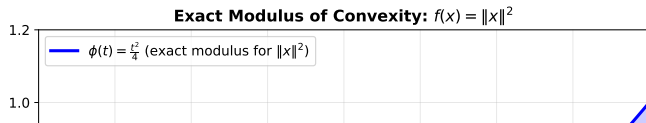
Definition

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and convex. The **exact modulus of convexity** is:

$$\varphi : \mathbb{R}_+ \rightarrow [0, +\infty]$$
$$t \mapsto \inf_{\substack{x, y \in \text{dom } f \\ \|x - y\| = t \\ \alpha \in [0, 1]}} \left(\frac{\alpha f(x) + (1 - \alpha)f(y) - f(\alpha x + (1 - \alpha)y)}{\alpha(1 - \alpha)} \right)$$

Interpretation:

- Measures the minimum “gap” between secant line and function
- Quantifies how much f curves away from linear interpolation
- $\varphi(0) = 0$ always (by definition)
- Larger φ means more curvature (stronger convexity)



Proposition 10.12: Properties of Modulus I

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and convex, with exact modulus φ . Then:

$$(\forall t \in \mathbb{R}_+)(\forall \gamma \in [1, +\infty[) \quad \varphi(\gamma t) \geq \gamma^2 \varphi(t).$$

Moreover, φ is increasing.

Proof strategy:

- 1 Establish that $\varphi(0) = 0$ and $\varphi(t) > 0$ for $t > 0$
- 2 For $\gamma \in]1, 2]$: Use specific choice of α and γ to show inequality
- 3 For $\gamma \in [2, +\infty[$: Write as product of factors in $]1, 2]$ and apply repeatedly
- 4 Monotonicity follows from definition

Corollary

f is uniformly convex if and only if φ vanishes only at 0; in this case, f is uniformly convex with modulus φ .

Corollary 10.13 & 10.14: Relationship to Moduli I

Corollary (10.13)

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and convex, with exact modulus φ . Then f is uniformly convex if and only if φ vanishes only at 0; in this case, f is uniformly convex with modulus φ .

[10.14] Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and convex with exact modulus φ , and define

$$\psi : \mathbb{R}_+ \rightarrow [0, +\infty] : t \mapsto \inf_{\substack{x, y \in \text{dom } f \\ \|x - y\| = t}} \left(\frac{1}{2}f(x) + \frac{1}{2}f(y) - f\left(\frac{1}{2}x + \frac{1}{2}y\right) \right)$$

Then: (i) $2\psi \leq \varphi \leq 4\psi$, and (ii) f is uniformly convex iff ψ vanishes only at 0.

Corollary 10.13 & 10.14: Relationship to Moduli II

Key relationship: The midpoint modulus ψ captures uniform convexity but is easier to compute than the full modulus φ .

Bounds interpretation:

- $\varphi \leq 4\psi$: The full gap is at most 4 times the midpoint gap
- $2\psi \leq \varphi$: The midpoint gap is at least half the full gap
- These bounds allow approximation of uniform convexity

Proposition 10.15: Powers of Uniformly Convex Functions

Let $g : \mathcal{H} \rightarrow \mathbb{R}_+$ be uniformly convex with exact modulus φ , and $p \in [1, +\infty[$. Then g^p is uniformly convex, with exact modulus χ satisfying:

$$\chi \geq 2^{1-2p} \min\{p^{2^{1-p}}, 1 - 2^{-p}\} \varphi^p.$$

Application to norms:

- For $g(x) = \|x\|$, which is uniformly convex with $\varphi(t) =$ determined by $\|\cdot\|$
- Then $\|x\|^p$ is also uniformly convex
- Example: $\|x\|_2^p$ is uniformly convex for all $p \in [1, +\infty[$
- Stronger convexity for $p > 2$

Takeaway: Powers of uniformly convex functions inherit uniform convexity with quantifiable bounds.

Example 10.16: Norm Powers are Uniformly Convex

Example

For $p \in [2, +\infty[$, the function $\|\cdot\|^p$ is uniformly convex.

Proof:

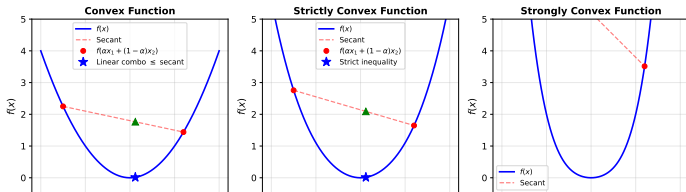
- By Example 10.9(ii), $\|\cdot\|^2$ is strongly convex with constant 2
- Hence $\|\cdot\|^2$ is uniformly convex
- Since $p/2 \in [1, +\infty[$, Proposition 10.15 gives that $(\|\cdot\|^2)^{p/2} = \|\cdot\|^p$ is uniformly convex

Partial order:

$p \in [1, 2) \Rightarrow$ not uniformly convex

$p = 2 \Rightarrow$ strongly convex

$p > 2 \Rightarrow$ uniformly convex (but not strongly)



Proposition 10.17: Uniform Convexity on Compact Sets

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper and convex, and let C be a nonempty compact convex subset of $\text{dom } f$ such that f is strictly convex on C and $f|_C$ is continuous. Then f is uniformly convex on C .

Significance:

- Compactness + strict convexity + continuity $\Rightarrow$ uniform convexity
- Useful for finite-dimensional optimization
- Shows that locally, smooth strictly convex functions behave uniformly

Example 10.19: Construction showing hypotheses matter

- $\mathcal{H} = \mathbb{R}^2$: $f(\xi, \eta) = \frac{\eta^2}{2\xi} + \xi^2$ (if $\xi > 0, \eta \in \mathbb{R}$)
- Is strictly convex on $C = B(0; \rho) \cap \text{dom } f$
- But NOT uniformly convex on C (fails at boundary)

Corollary 10.18: Finite-Dimensional Case

Corollary

Suppose $\mathcal{H}$ is finite-dimensional, let $f : \mathcal{H} \rightarrow \mathbb{R}$ be strictly convex, and let C be a nonempty bounded closed convex subset of $\mathcal{H}$. Then f is uniformly convex on C .

Proof: Combine Corollary 8.40 (continuity in finite dimensions) and Proposition 10.17.

Practical implications:

- In finite dimensions: Strict convexity + compactness $\Rightarrow$ Uniform convexity
- No need to verify the modulus directly
- Automatic local uniform convexity near interior of domain
- Important for convergence analysis of optimization algorithms

Definition 10.20: Quasiconvex Functions

Definition

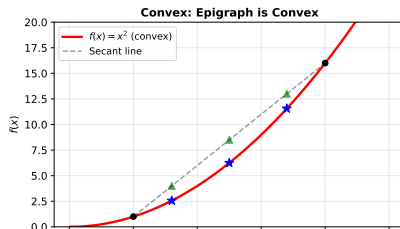
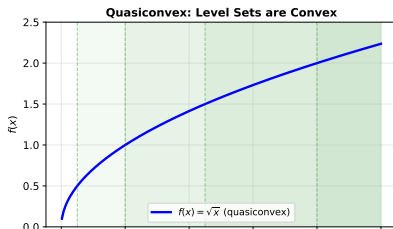
Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$. Then f is **quasiconvex** if its lower level sets $(\text{lev}_\xi f)_{\xi \in \mathbb{R}}$ are convex.

Equivalently:

$$(\forall x, y \in \text{dom } f)(\forall \alpha \in [0, 1]) \quad f(\alpha x + (1 - \alpha)y) \leq \max\{f(x), f(y)\}$$

Key observations:

- Quasiconvexity is WEAKER than convexity
- Convex $\Rightarrow$ Quasiconvex (converse false)
- Level sets are more fundamental than epigraph
- More flexible than convexity for applications



Examples 10.21-10.23: Quasiconvex Functions

Example (10.21)

Every convex function is quasiconvex (direct from Corollary 8.5).

Example (10.22)

Every increasing or decreasing function $f : \mathbb{R} \rightarrow]-\infty, +\infty]$ is quasiconvex.

Example (10.23)

If $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ is quasiconvex and $\eta \in \mathbb{R}$, then $\min\{f, \eta\}$ is quasiconvex.

Proof (10.23): For $g = \min\{f, \eta\}$, if $\xi \leq \eta$:

$$\text{lev}_\xi g = \text{lev}_\xi f \cap \text{lev}_\xi \eta = \text{lev}_\xi f$$

which is convex (as f is quasiconvex).

Propositions 10.24-10.26: Operations on Quasiconvex Functions

[10.24] Let $(f_i)_{i \in I}$ be a family of quasiconvex functions from $\mathcal{H}$ to $] - \infty, +\infty]$. Then $\sup_{i \in I} f_i$ is quasiconvex.

[10.25] Let $f : \mathcal{H} \rightarrow] - \infty, +\infty]$ be quasiconvex. The following are equivalent:

- ① f is weakly sequentially lower semicontinuous
- ② f is sequentially lower semicontinuous
- ③ f is lower semicontinuous
- ④ f is weakly lower semicontinuous

[10.26] $f : \mathcal{H} \rightarrow] - \infty, +\infty]$ is quasiconvex if and only if

$$(\forall x, y \in \text{dom } f)(\forall \alpha \in [0, 1]) \quad f(\alpha x + (1 - \alpha)y) \leq \max\{f(x), f(y)\}$$

Definition 10.27: Strict and Uniform Quasiconvexity

Definition

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper, $C \subseteq \text{dom } f$ nonempty, and $\phi : \mathbb{R}_+ \rightarrow [0, +\infty]$ increasing and vanishing only at 0. Then f is:

- 1 **Strictly quasiconvex** if

$$(x \neq y, \alpha \in (0, 1)) \Rightarrow f(\alpha x + (1 - \alpha)y) < \max\{f(x), f(y)\}$$

- 2 **Uniformly quasiconvex** with modulus ϕ if

$$f(\alpha x + (1 - \alpha)y) + \alpha(1 - \alpha)\phi(\|x - y\|) \leq \max\{f(x), f(y)\}$$

- 3 Strict and uniform versions restricted to C (on C)

Hierarchy:

Uniformly quasiconvex $\Rightarrow$ Strictly quasiconvex $\Rightarrow$ Quasiconvex

Examples 10.29-10.32: Illustrating Variants

Example (10.29)

A strictly increasing or strictly decreasing proper $f : \mathbb{R} \rightarrow]-\infty, +\infty]$ is strictly quasiconvex.

Example (10.30)

Let $f : \mathcal{H} \rightarrow \mathbb{R}$ and $\phi : \text{ran } f \rightarrow \mathbb{R}$ increasing.

- 1 If f strictly quasiconvex and ϕ strictly increasing, then $\phi \circ f$ strictly quasiconvex
- 2 If $\phi \circ f$ strictly convex, then f strictly quasiconvex

```
1 # Numerical illustration: Composing functions preserves quasiconvexity
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 x = np.linspace(0.1, 5, 100)
6 f = np.sqrt(x) # quasiconvex (strictly increasing)
7 phi = lambda y: np.log(y) # strictly increasing
8 phi_of_f = phi(f) # strictly quasiconvex
9
10 plt.plot(x, phi_of_f, 'b-')
11 plt.title(r'$ \phi(f(x)) = \log(\sqrt{x})$ is strictly quasiconvex')
```

Examples 10.31-10.32: Subtle Distinctions

Example (10.31)

For $p \in \mathbb{R}_{++}$:

- 1 $\|\cdot\|^p$ is strictly quasiconvex
- 2 If $\mathcal{H} \neq \{0\}$ and $p < 1$, then $\|\cdot\|^p$ is NOT convex and NOT uniformly quasiconvex

Example (10.32)

$f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \nu x$ where $\nu \in \mathbb{R}_{++}$ is:

- Convex
- NOT strictly convex (affine!)
- Uniformly quasiconvex with modulus $\phi(t) = \nu t$

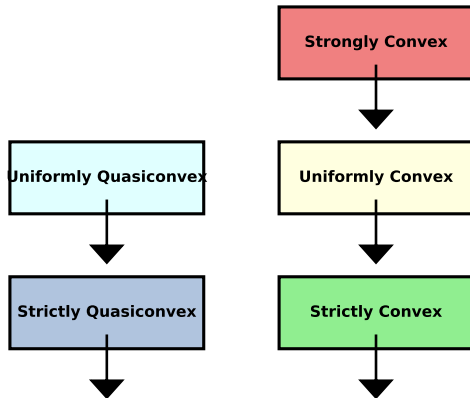
Takeaway:

- Quasiconvexity variants decouple different notions of convexity
- Allows modeling functions with monotone behavior
- Useful in economics, non-convex optimization, and games

Hierarchy of Convexity Classes

Hierarchy of Convexity Classes

Strong implications (top to bottom)



Summary of Key Results

Uniform Convexity (Section 10.2):

- Quantifies how much function curves away from secant lines
- Exact modulus $\varphi(t)$ measures minimal gap
- Strong convexity: $\varphi(t) = (\beta/2)t^2$ (quadratic gap)
- Equivalent to $f - (\beta/2)\|\cdot\|^2$ being convex
- In finite dimensions: Strict convexity + compactness $\Rightarrow$ Uniform convexity

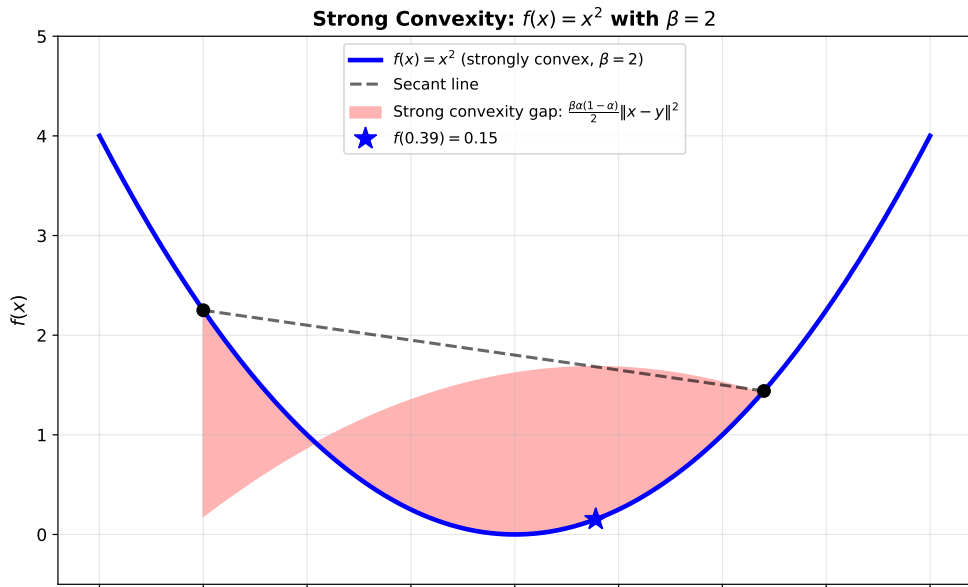
Quasiconvexity (Section 10.3):

- Defined by convexity of level sets (not epigraph)
- Weaker than convexity: all convex are quasiconvex, but not vice versa
- Preserved under: supremum, lower semicontinuity
- Strict variant: strict inequality for distinct points
- Uniform variant: adds modulus term for gap control

Applications:

- Uniform convexity $\Rightarrow$ Unique minimizers, fast convergence
- Quasiconvexity $\Rightarrow$ Non-convex but tractable optimization
- Monotone and level-set based problems

Computational Example: Strong Convexity



Python Implementation: Computing Moduli

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 def exact_modulus_estimation(f, x1, x2, t, num_alpha=50):
5     """Estimate exact modulus  $\phi(t)$  for convex function  $f$ """
6     alphas = np.linspace(0, 1, num_alpha)
7     gaps = []
8
9     for alpha in alphas:
10         x_mid = alpha * x1 + (1 - alpha) * x2
11
12         # Gap = (af(x1) + (1-a)f(x2) - f(a*x1 + (1-a)*x2)) / (a(1-a))
13         if alpha > 0 and alpha < 1:
14             numerator = (alpha * f(x1) + (1 - alpha) * f(x2)
15                          - f(x_mid))
16             gap = numerator / (alpha * (1 - alpha))
17             gaps.append(gap)
18
19     #  $\phi(t) = \inf$  over gaps
20     return np.min(gaps) if gaps else np.inf
```

Exercises: Key Problems

Core exercises (10.1-10.10):

- 10.3: Show f sublinear iff $\text{epi } f$ is cone
- 10.4: Verify Example 10.9
- 10.5: Show $\|\cdot\|^p$ not uniformly convex for $p < 1$ (dimension ≥ 2)
- 10.8: Compute modulus ψ for $f(x) = |x|^4$
- 10.10: Strict convexity + boundedness $\Rightarrow$ uniform convexity

Quasiconvexity exercises (10.11-10.18):

- 10.12: Non-convex but uniformly quasiconvex examples
- 10.13: Function strictly quasiconvex but not convex
- 10.15: Sum of quasiconvex + convex may not be quasiconvex
- 10.16: Marginal function inheritance of quasiconvexity

Difficulty scaling: 10.1-10.8 (foundational) $\rightarrow$ 10.9-10.18 (refinements)

Open Research Directions

- **Quantitative rates:** Compute explicit moduli for structured problems
 - ℓ_p spaces with various p
 - Matrix-valued functions (spectral norms)
 - PDE-constrained convex optimization
- **Quasiconvex optimization:** Algorithms for non-convex but quasiconvex problems
 - Feasible direction methods
 - Level-set based approaches
 - Connections to DC and difference-of-convex programming
- **Composite structures:**
 - Preserve uniform convexity under composition/convolution
 - Generalizations to non-Euclidean geometries
 - Bregman divergences and mirror descent
- **Applications:**
 - Faster convergence guarantees in machine learning
 - Quasiconvex games and auction design
 - Monotone operators with strong convexity structure

Key Takeaways

- Uniform and strong convexity quantify curvature via moduli
- Strong convexity: $f - (\beta/2)\|\cdot\|^2$ is convex
- Quasiconvexity: weaker notion based on level sets
- Hierarchy: Strong $\Rightarrow$ Uniform $\Rightarrow$ Strict $\Rightarrow$ Convex
- Finite-dimensional: Strict + compact $\Rightarrow$ Uniform
- Quasiconvex $\not\Rightarrow$ Convex but Convex $\Rightarrow$ Quasiconvex
- Applications: Optimization convergence, monotone operators, games